

Music Player App

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The purpose of this project was to design and develop a music application that allows users to act as a regular music player, edit a song's tempo and pitch and allow the changing of the theme.

The problem addressed by this project is that many existing music applications do not have the ability to change the pitch or tempo of a song separately.

This application was created to demonstrate both technical skills and the ability to design a functional, user-centered product.

Before developing the application, research was conducted on existing music applications and mobile development tools.

Applications such as default music players, spotify, pandora, itunes and more were analyzed to understand common features like playlists, playback controls, and user interfaces.

In addition, research into Android development using Kotlin and Jetpack Compose was completed to understand how to build responsive user interfaces and manage media playback.

This background research helped guide the design and functionality of the final application.

The project was developed using the following technologies:

- Programming Language(s): Kotlin
- Frameworks/Libraries: Jetpack Compose, ExoPlayer (Media3)
- Development Environment: Android Studio
- Hardware: Android device/emulator for testing

These tools were selected due to their compatibility with Android development and ability to support media playback features.

The project was required to include the following features:

- Ability to load and play music files
- Playback controls (play, pause, skip)
- Playlist functionality

- User interface for navigation
- Tempo alteration
- Pitch alteration
- Repeat and shuffle options
- Theme changing abilities

The application was designed to meet these requirements while maintaining usability and functionality.

The application was implemented using a structured design consisting of multiple screens, including a song screen, playlist screen, theme screen, and music player screen.

The music playback functionality was handled using ExoPlayer, which allowed for efficient audio streaming and control.

Fig 1. Song Screen - displays the title of the app while in dark mode and with a pink border selection displaying regular music player controls like play, skip, order, repeat, shuffle, and a time stamp slider. The three dots in the top right corner lead to the drop down shown in fig 2.

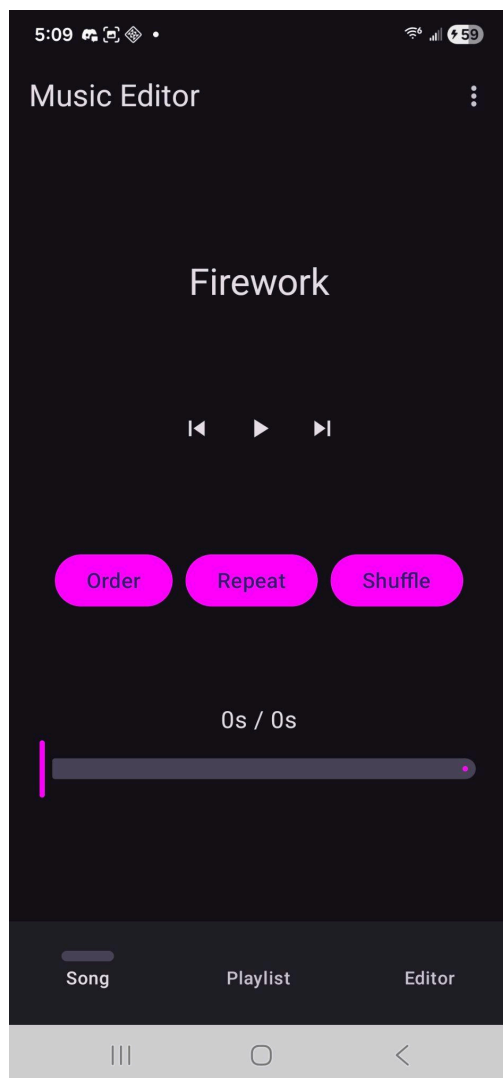


Fig 2. Drop down selection for theme and currently non functional help option

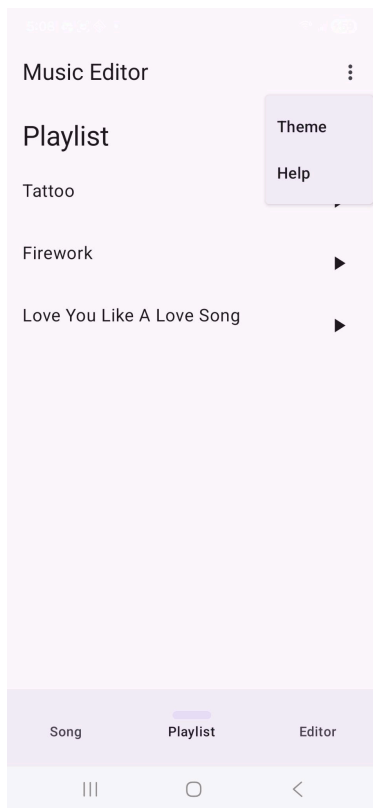


Fig 3. Music Editor Screen - Shows movable sliders with max values on either end to adjust the tempo and pitch separately along with reset buttons to turn the edits back to normal.

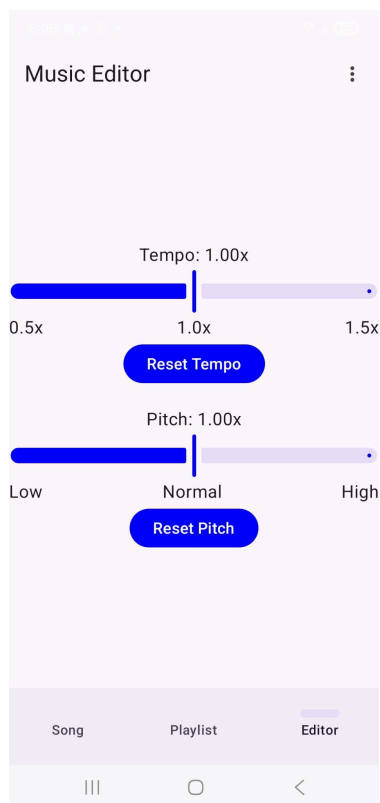


Fig 4. Playlist Screen - this screen displays the songs listed on the device along with a play button that would redirect the user to the song menu with the song they picked.

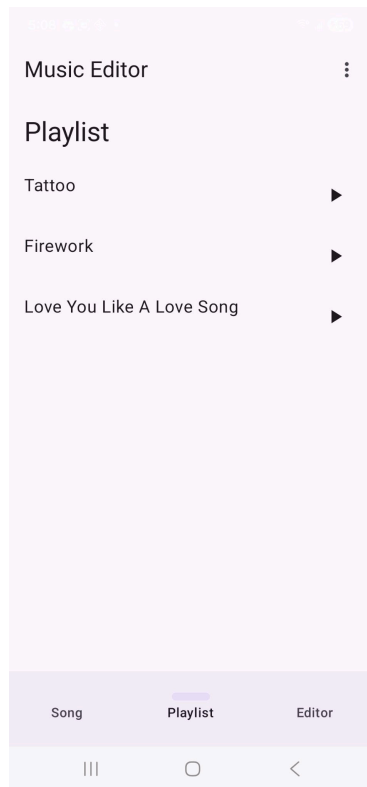


Fig 4. Theme Screen - this is the screen that comes from the drop down and you can used the back arrow from the phone to go back to the main set of screens. There is a single sliding button for dark mode or light mode if its off and the border colors are the current options available with pink being the current selection.

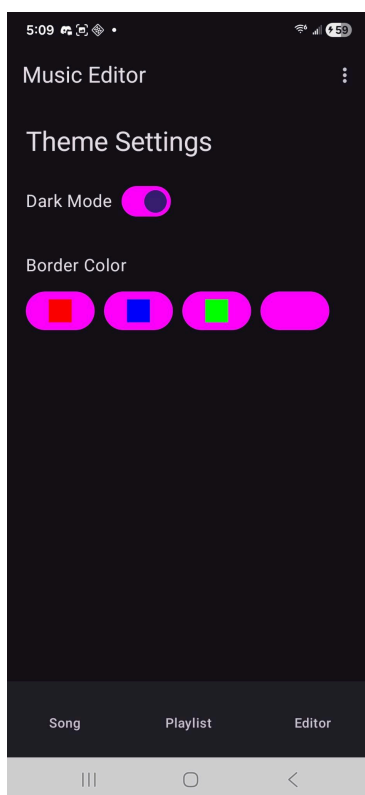
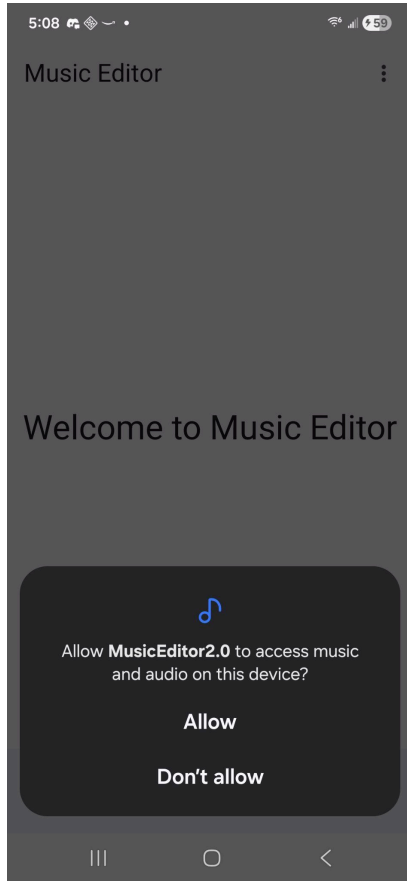


Fig. 5 The permission over the default screen. The permission is asking at the start of the app to be able to access the users songs. Then the default screen in the background goes straight to the playlist screen once tapped.



Each screen was designed using Jetpack Compose, allowing for a modern and responsive user interface.

Source Code Repository: <https://github.com/dixkay5181/MusicEditor2.0v1>

The application was tested to ensure all features functioned correctly.

Testing included:

- Verifying music playback functionality
- Testing user interface navigation
- Ensuring controls such as play, pause, and skip worked properly
- Testing additional features like [tempo/pitch adjustments]

Both emulator and physical device testing were used.

The test results showed that the application successfully performed its main intended functions.

Most features operated as expected, including music playback and user interface navigation. Some minor issues were identified and addressed during development.

Overall, the application met the primary functional requirements.

Several challenges were encountered during development.

One challenge was implementing audio playback using ExoPlayer and ensuring it worked correctly with the user interface which caused a massive crash that needed the entire app to be rebuilt.

Another challenge was managing application state across different screens.

These challenges were overcome through research, testing, and iterative improvements.

Future improvements to the application may include:

- Enhanced user interface design
- Additional audio editing features
- Improved playlist management
- The addition of a help function
- Ability to download edited songs

These enhancements would further improve usability and functionality.

The presentation slides used during the defense are included below.

<https://github.com/dixkay5181/CSU-Senior-Project/blob/master/docs/Music%20Player.pdf>